

TASK 3 CAUSALITY ASSESSMENT

Detailed Case Studies & Clinical Reasoning Using the WHO-UMC Scale

Assessment Method

WHO-UMC Standardised Case Causality Assessment System

Training Cases

Case 1 — Certain / Definite

Case 2 — Probable / Likely

Case 3 — Possible

All cases in this document are fictional educational scenarios created for pharmacovigilance internship practice.

1. Learning Objectives

The purpose of this assignment is to demonstrate how a pharmacovigilance professional evaluates the relationship between a suspect medicinal product and an adverse event.

Causality assessment should be systematic, evidence-based and appropriately cautious. The presence of an event after drug exposure does not automatically establish that the drug caused the event.

This exercise focuses on temporal relationship, dechallenge, rechallenge, alternative causes, known reaction profile, patient factors, objective evidence and clinical reasoning, assessed exclusively against the WHO-UMC causality categories.

2. WHO-UMC Causality Assessment Scale

The WHO-UMC system is a standardised, criteria-based scale used by pharmacovigilance centres worldwide to classify the likelihood that a medicine caused a reported adverse event. It comprises six categories, ranging from Certain to Unassessable/Unclassifiable.

Category	Defining Criteria
Certain	Event with a plausible time relationship to drug intake; cannot be explained by concurrent disease or other drugs; response to withdrawal is clinically plausible; event is pharmacologically or phenomenologically definitive; satisfactory rechallenge, if necessary.
Probable / Likely	Event with a reasonable time relationship to drug intake; unlikely to be attributed to concurrent disease or other drugs; clinically reasonable response to withdrawal; rechallenge information not required.
Possible	Event with a reasonable time relationship to drug intake; could also be explained by concurrent disease or other drugs; information on drug withdrawal may be lacking or unclear.

Unlikely	Event with a time relationship that makes a causal connection improbable; disease or other drugs provide a plausible alternative explanation.
Conditional / Unclassified	Event reported as an adverse reaction, but more data are essential for a proper assessment, or additional data are under examination.
Unassessable / Unclassifiable	A report suggesting an adverse reaction that cannot be judged because information is insufficient or contradictory, and cannot be supplemented or verified.

3. Evidence Checklist

Factor	Assessment Question	Why It Matters
Temporal relationship	Did the event occur after drug exposure and within a plausible interval?	Supports causality when compatible, but timing alone does not prove causation.
Dechallenge	Did the event improve after withdrawal?	A clinically plausible improvement can strengthen the association.
Rechallenge	Did the event recur after re-exposure?	A positive rechallenge can provide strong supporting evidence.

Factor	Assessment Question	Why It Matters
Alternative causes	Could disease, another medicine or another exposure explain the event?	A credible alternative weakens the drug-event relationship.
Known reaction	Is the event consistent with known pharmacology or clinical experience?	Supports biological and clinical plausibility.
Objective evidence	Is there laboratory, examination, imaging or other objective evidence?	Can strengthen confidence in event characterization.

CASE 1 — CERTAIN / DEFINITE

A. Patient Profile & Background

Patient: A.K., 28-year-old female, 60 kg.

Indication: Acute bacterial sinusitis.

Suspect medicine: Amoxicillin 500 mg orally.

Medical background: No previous drug allergy or similar reaction is documented. No new food, cosmetic, insect exposure or competing suspect medicine is reported.

B. Detailed Chronology

Day 0, 09:00: Amoxicillin is administered for bacterial sinusitis.

09:10: The patient develops generalized urticaria and intense itching. Lip and facial swelling then appear.

09:15: Wheezing and hypotension develop. The suspect medicine is immediately stopped and emergency treatment is initiated.

09:45: Respiratory and cardiovascular findings improve substantially after treatment.

Later clinical observation: An inadvertent re-exposure to the same medicine occurs and the same acute clinical pattern recurs. The medicine is subsequently avoided.

C. Adverse Reaction & Clinical Management

Reaction: Generalized urticaria, angioedema, wheezing and hypotension.

Dechallenge: Compatible improvement after withdrawal and emergency treatment.

Rechallenge: Positive, with recurrence of the same clinical pattern after re-exposure.

Outcome: Recovery after treatment and permanent avoidance of the suspect medicine.

D. WHO-UMC Evidence Analysis

The temporal relationship is exceptionally strong because the event begins within minutes of administration. The clinical phenotype is highly compatible with an immediate hypersensitivity/anaphylactic-type reaction. No reasonable competing cause is identified in this fictional scenario. Improvement follows withdrawal and treatment, providing compatible dechallenge evidence. Most importantly, the same reaction recurs following re-exposure, providing strong additional support. Under the WHO-UMC framework, a Certain assessment requires more than chronology alone: the event should be definitive pharmacologically or phenomenologically and should not be adequately explained by disease or another medicine. All five defining criteria for the Certain category are satisfied in this case.

E. WHO-UMC Criterion-by-Criterion Assessment

Criterion	WHO-UMC Requirement	Case Evidence	Met?
Time relationship	Plausible time relationship to drug intake.	Onset within 10 minutes of administration.	Yes
Alternative causes	Cannot be explained by concurrent disease or another drug.	No competing medicine, food or exposure identified.	Yes

Dechallenge	Clinically plausible response to withdrawal.	Marked improvement following withdrawal and treatment.	Yes
Criterion	WHO-UMC Requirement	Case Evidence	Met?
Rechallenge	Satisfactory rechallenge, if necessary.	Same clinical pattern recurred on inadvertent re-exposure.	Yes
Event definitive	Pharmacologically or phenomenologically definitive event.	Classic immediate hypersensitivity/anaphylactic phenotype.	Yes

Final WHO-UMC Causality Classification: Certain / Definite

F. Detailed Conclusion

The temporal relationship is exceptionally strong because the event begins within minutes of administration. The clinical phenotype is highly compatible with an immediate hypersensitivity/anaphylactic-type reaction. No reasonable competing cause is identified in this fictional scenario. Improvement follows withdrawal and treatment, providing compatible dechallenge evidence. Most importantly, the same reaction recurs following re-exposure, providing strong additional support. Under the WHO-UMC framework, a Certain assessment requires more than chronology alone: the event should be definitive pharmacologically or phenomenologically and should not be adequately explained by disease or another medicine. All five defining criteria for the Certain category are satisfied in this case.

G. Pharmacovigilance Learning Point

This case demonstrates convergence of multiple strong evidence streams: exceptionally close timing, a recognized clinical phenotype, absence of an identified alternative cause, compatible improvement following withdrawal and positive rechallenge. The positive rechallenge is particularly informative. However, in real pharmacovigilance practice, a patient should not be deliberately re-exposed simply to strengthen causality evidence. Patient safety takes priority.

CASE 2 — PROBABLE / LIKELY

A. Patient Profile & Background

Patient: R.S., 34-year-old female, 62 kg.

Indication: Acute bacterial sinusitis.

Suspect medicine: Amoxicillin 500 mg orally three times daily.

Concomitant medicines: None reported.

Background: No previous similar reaction is documented.

B. Detailed Chronology

Day 1: Amoxicillin therapy is initiated.

Day 2: Generalized itching and a maculopapular rash develop over the trunk and upper limbs. No new medicine, food, cosmetic or obvious competing exposure is reported.

Day 2: Amoxicillin is discontinued and symptomatic antihistamine treatment is given.

Days 3–5: The rash and itching gradually resolve.

Day 5: The patient has recovered. No rechallenge is performed.

C. Adverse Reaction & Clinical Management

Reaction: Generalized pruritic maculopapular rash.

Dechallenge: Positive/compatible because the reaction improves after withdrawal.

Rechallenge: Not performed.

Outcome: Recovered.

D. WHO-UMC Evidence Analysis

The temporal relationship is reasonable because the reaction develops during treatment. The event improves after withdrawal, providing supportive dechallenge evidence. No clear competing medicine or exposure is reported. However, there is no positive rechallenge and the event does not have the unusually definitive evidence seen in Case 1. Under WHO-UMC, rechallenge is not required for a Probable/Likely classification. The overall evidence therefore supports a stronger association than Possible, while remaining below Certain.

E. WHO-UMC Criterion-by-Criterion Assessment

Criterion	WHO-UMC Requirement	Case Evidence	Met?
Time relationship	Reasonable time relationship to drug intake.	Rash developed on Day 2 of therapy.	Yes
Alternative causes	Unlikely to be attributed to concurrent disease or other drug.	No concomitant medicine or new exposure reported.	Yes
Dechallenge	Clinically reasonable response to withdrawal.	Rash and itching gradually resolved after discontinuation.	Yes

Rechallenge	Not required for this category.	No rechallenge performed.	N/A
Criterion	WHO-UMC Requirement	Case Evidence	Met?
Event definitive	Event pattern consistent with a recognized reaction, though not as unequivocal as Certain.	Maculopapular rash is a recognized amoxicillin reaction.	Partial

Final WHO-UMC Causality Classification: Probable / Likely

F. Detailed Conclusion

The temporal relationship is reasonable because the reaction develops during treatment. The event improves after withdrawal, providing supportive dechallenge evidence. No clear competing medicine or exposure is reported. However, there is no positive rechallenge and the event does not have the unusually definitive evidence seen in Case 1. Under WHO-UMC, rechallenge is not required for a Probable/Likely classification. The overall evidence therefore supports a stronger association than Possible, while remaining below Certain.

G. Pharmacovigilance Learning Point

This case represents a common pharmacovigilance scenario in which a suspected ADR has a reasonable chronology and improves after the suspect medicine is withdrawn, but re-exposure evidence is unavailable. A Probable/Likely classification appropriately communicates meaningful evidence without overstating certainty. The assessor should document the evidence available in the case rather than attempting to create additional evidence through an unnecessary rechallenge.

CASE 3 — POSSIBLE

A. Patient Profile & Background

Patient: M.P., 45-year-old male, 74 kg.

Indication: Hypertension.

Suspect medicine: Drug X, a fictional antihypertensive, 10 mg once daily.

Relevant circumstances: During the same period, the patient sleeps only four to five hours per night and drinks substantially less fluid than usual.

B. Detailed Chronology

Day 1: Drug X is started.

Days 1–3: Poor sleep and reduced fluid intake continue.

Day 3: Mild-to-moderate headache develops. Drug X is continued.

Day 4: The patient increases fluid intake and rests adequately. The headache resolves later that day.

No dechallenge or rechallenge is performed.

C. Adverse Reaction & Clinical Management

Reaction: Headache.

Dechallenge: Cannot be evaluated because Drug X is continued.

Rechallenge: Not performed.

Alternative causes: Dehydration and sleep deprivation are credible explanations.

Outcome: Headache resolves while Drug X remains in use.

D. WHO-UMC Evidence Analysis

The event occurs after the medicine is started, so the temporal relationship is compatible. However, the symptom is nonspecific and there are credible alternative explanations. Dehydration and sleep deprivation could independently produce headache. The medicine is not withdrawn, so dechallenge cannot be assessed. The symptom resolves after hydration and rest while Drug X continues. These factors substantially limit the strength of the drug-event relationship. A Possible classification therefore reflects the available evidence without overstating causality.

E. WHO-UMC Criterion-by-Criterion Assessment

Criterion	WHO-UMC Requirement	Case Evidence	Met?
Time relationship	Reasonable time relationship to drug intake.	Headache developed on Day 3 of therapy.	Yes
Alternative causes	Could also be explained by concurrent disease or other factors.	Dehydration and sleep deprivation are credible explanations.	Yes (alt. cause present)
Dechallenge	Withdrawal information may be lacking or unclear.	Drug X was continued; dechallenge not assessed.	Not evaluable

Rechallenge	Not required for this category.	No rechallenge performed.	N/A
Criterion	WHO-UMC Requirement	Case Evidence	Met?
Event definitive	Event need not be definitive for this category.	Headache is a nonspecific symptom with multiple possible causes.	No

Final WHO-UMC Causality Classification: Possible

F. Detailed Conclusion

The event occurs after the medicine is started, so the temporal relationship is compatible. However, the symptom is nonspecific and there are credible alternative explanations. Dehydration and sleep deprivation could independently produce headache. The medicine is not withdrawn, so dechallenge cannot be assessed. The symptom resolves after hydration and rest while Drug X continues. These factors substantially limit the strength of the drug-event relationship. A Possible classification therefore reflects the available evidence without overstating causality.

G. Pharmacovigilance Learning Point

This case demonstrates why temporal association alone should not be interpreted as proof of causality. The headache occurred after drug initiation, but credible alternative causes were present. There was no dechallenge, no rechallenge and no objective evidence establishing drug causation. The symptom also resolved while the suspect medicine was continued. The appropriate pharmacovigilance approach is therefore to document the uncertainty and classify the relationship as Possible.

4. Comparative Interpretation

The three cases intentionally demonstrate different levels of evidentiary strength. The WHO-UMC category assigned should reflect the evidence actually documented in the case.

Evidence	Case 1	Case 2	Case 3
Temporal relationship	Very strong; minutes	Reasonable; 2 days	Reasonable; 3 days
Dechallenge	Compatible improvement	Improvement after withdrawal	Unavailable
Rechallenge	Positive	Not performed	Not performed
Alternative causes	None identified	No clear alternative	Credible alternatives
WHO-UMC category	Certain	Probable/Likely	Possible

5. Practical Pharmacovigilance Takeaways

- 1. Build the chronology first.** Record drug start, dose, event onset, treatment, withdrawal, recovery and any re-exposure information.
- 2. Do not equate 'after the drug' with 'because of the drug'.** Temporal association supports an assessment but does not independently establish causation.
- 3. Search actively for alternative causes.** Disease, concomitant medicines and other exposures can materially change the assessment.
- 4. Interpret dechallenge and rechallenge carefully.** They are evidence that must be considered in clinical context.
- 5. Never invent missing information.** If a criterion cannot be evaluated from the available case information, document it as unknown or not available rather than assuming an answer.
- 6. Assess every case against all six WHO-UMC categories.** Even when a case clearly fits one category, briefly checking it against the others confirms the classification is well-reasoned rather than assumed.

6. Final Conclusion

The three cases demonstrate a progression in evidentiary strength under the WHO-UMC scale. **Case 1 — Certain/Definite** satisfies all defining criteria, including a positive rechallenge. **Case 2 — Probable/Likely** shows reasonable chronology, compatible dechallenge and no clear alternative explanation, without rechallenge. **Case 3 — Possible** has a compatible temporal relationship but credible alternative explanations and absent dechallenge information.